

VNUHCM - University of Science
Faculty of Information Technology

CS486 – Lecture 4

Query Languages

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Content

- Introduction
- Define relation schemas
- Manipulate on relation schemas
 - Data modification
 - Data retrieval
 - Basic query
 - * On one schemas
 - * On two or many schemas
 - **Advanced query**
- Summary

Relational algebra

- Selection
- Projection
- Cartesian product
- Join operation
- **Set operations**
- Division operation
- Other operations
- Modification operations

Set operation

- Relation is a set of tuples
 - The union $R \cup S$
 - The intersection $R \cap S$
 - The difference $R - S$
- Union Compatibility
 - Two relation schemas $R(A_1, A_2, \dots, A_n)$ and $S(B_1, B_2, \dots, B_n)$ are union compatibility if
 - The same degree n
 - And $DOM(A_i) = DOM(B_i)$, $1 \leq i \leq n$
- The result of $\cup$, $\cap$, and $-$ operations
 - Relation

Union

- Given two relations R & S that are union compatible
- The union of R and S
 - Notation $R \cup S$
 - A relation consists of tuples that are in R or S or both (an element appears only one)

$$R \cup S = \{t / t \in R \vee t \in S\}$$

- Example

R	A	B
	α	1
	α	2
	β	1

S	A	B
	α	2
	β	3

$R \cup S$	A	B
	α	1
	α	2
	β	1
	α	2
	β	3

Intersection

- Given two relations R & S that are union compatible
- The intersection of R and S
 - Denotation $R \cap S$
 - A relation consists of tuples that are in R and S

$$R \cap S = \{ t / t \in R \wedge t \in S \}$$

- Example

R	A	B
	α	1
	α	2
	β	1

S	A	B
	α	2
	β	3

$R \cap S$	A	B
	α	2

Difference

- Given two relations R & S that are union compatible
- The difference of R and S
 - Denotation $R - S$
 - A relation consists of tuples that are in R but not in S

$$R - S = \{ t / t \in R \wedge t \notin S \}$$

- Example

R	A	B
	α	1
	α	2
	β	1

S	A	B
	α	2
	β	3

R - S	A	B
	α	1
	β	1

Properties

- Commutative

$$R \cup S = S \cup R$$

$$R \cap S = S \cap R$$

- Associative

$$R \cup (S \cup T) = (R \cup S) \cup T$$

$$R \cap (S \cap T) = (R \cap S) \cap T$$

The complete set

- The set of relational algebra operations

$\{\sigma, \pi, \times, -, \cup\}$ is called a complete set

- Any of other relational algebra operations can be expressed as a sequence of operations from this set

- Example

- $R \cap S = R - (R - S)$
- $R \bowtie_c S = \sigma_c(R \times S)$

Example 17

- Find the SSN of employees who work on projects or have dependents

Example 18

- Find the SSN of employees who work on projects and have dependents

Example 19

- Find the SSN of employees who do not have any dependents

Nested query

```
SELECT SSN, FName  
FROM EMPLOYEE, DEPT_LOCATIONS  
WHERE DLocation= 'Houston' AND DNo=DNumber
```

Outer query

```
SELECT <List_of_columns>  
FROM <List_of_tables>  
WHERE <Set_comparision> (  
    SELECT <List_of_columns>  
    FROM <List_of_tables>  
    WHERE <Condition>)
```

Subquery

Set operations in SQL

- SQL has implemented set operators
 - UNION
 - INTERSECT
 - EXCEPT

- The result is a set
 - Eliminate identical tuples
 - To keep identical tuples
 - UNION ALL
 - INTERSECT ALL
 - EXCEPT ALL

Set operations in SQL

■ Syntax

```
SELECT <Column_list> FROM <Table_list> WHERE <Condition>
```

UNION [ALL]

```
SELECT <Column_list> FROM <Table_list> WHERE <Condition>
```

```
SELECT <Column_list> FROM <Table_list> WHERE <Condition>
```

INTERSECT [ALL]

```
SELECT <Column_list> FROM <Table_list> WHERE <Condition>
```

```
SELECT <Column_list> FROM <Table_list> WHERE <Condition>
```

EXCEPT [ALL]

```
SELECT <Column_list> FROM <Table_list> WHERE <Condition>
```

Example 20

- Find the project numbers that have
 - Employees with the last name “Wong”,
 - Or been controlled by the department whose manager has the last name “Wong”

Example 21

- Find employees who have dependents with the same first name and sex as the employees

Example 21'

- Find employees whose first name and sex are the same to dependents

Example 19

- Find the SSN of employees who do not have any dependents

Nested query

- Queries can have several nested levels
 - Usually three
- Subqueries of a WHERE clause are connected by logical connective
 - OR, AND
- Subqueries will return
 - A single attribute and a single tuple (a single value)
 - A table (a set or multiset of tuples)

Nested query

- WHERE clause of the outer query
 - <Expression> <set operation> <subquery>
 - Set comparison includes many operators
 - IN, NOT IN
 - ALL
 - ANY or SOME
 - Check whether the result of subqueries is empty or not
 - EXISTS
 - NOT EXISTS

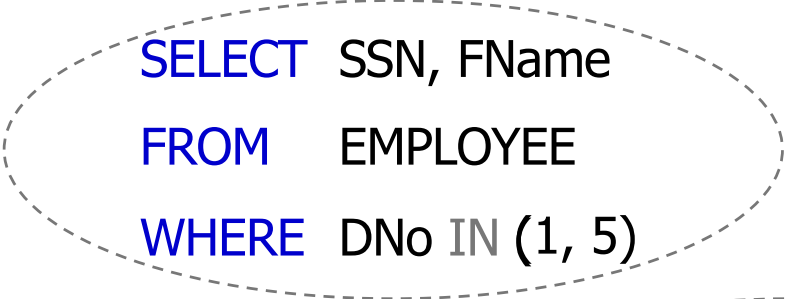
Nested query

■ Categories

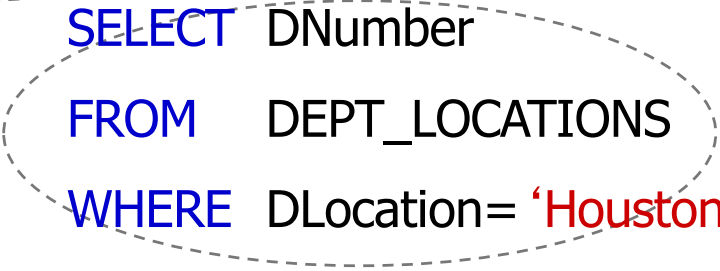
- Subqueries that produce scalar values
 - WHERE clause of subqueries do not refer to attributes of relations in FROM clause of the outer query
 - Subqueries will be performed before the outer query, and be executed just one time
- Correlated subqueries
 - WHERE clause of subqueries refer to at least one attribute of relations in FROM clause of the outer query
 - Subqueries will be executed many times, each time will correlate to one tuple of the outer query

Example – Scalar value subquery

```
SELECT SSN, FName  
FROM EMPLOYEE, DEPT_LOCATIONS  
WHERE DLocation= 'Houston' AND DNo=DNumber
```



```
SELECT SSN, FName  
FROM EMPLOYEE  
WHERE DNo IN (1, 5)
```



```
SELECT DNumber  
FROM DEPT_LOCATIONS  
WHERE DLocation= 'Houston' )
```

Example 20

```
SELECT WO.PNo
FROM   EMPLOYEE EM, WORKS_ON WO
WHERE  EM.SSN=WO.ESSN AND EM.LName= 'Wong'
UNION
SELECT PR.PNumber
FROM   EMPLOYEE EM, DEPARTMENT DE, PROJECT PR
WHERE  EM.SSN=DE.MgrSSN AND DE.DNumber=PR.DNum
AND EM.LName= 'Wong'
```


Example 20

SELECT DISTINCT PName

FROM PROJECT

WHERE PNumber IN (

SELECT PNo

FROM EMPLOYEE, WORKS_ON

WHERE SSN=ESSN AND LName= 'Wong')

OR PNumber IN (

SELECT PNumber

FROM EMPLOYEE, DEPARTMENT, PROJECT

WHERE SSN=ESSN AND PNumber=DNum

AND LName= 'Wong')

Example 19

- Find employees who have no dependents

Example 22

- Find employees whose salary is greater than at least one salary of employees in the department 4

Example 23

- Find employees whose salary is greater than all salaries of employees in the department 4

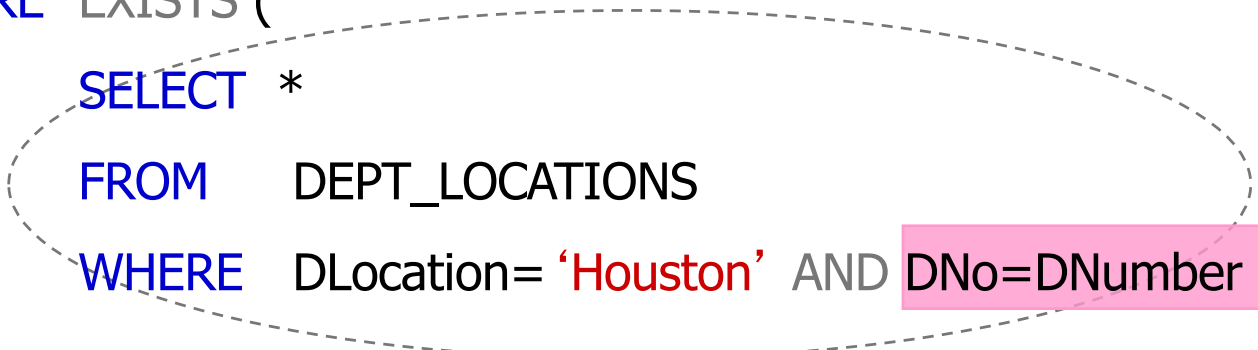
Example 24

- Find managers who have at least one dependent

Example - Correlated subquery

```
SELECT SSN, FName  
FROM EMPLOYEE, DEPT_LOCATIONS  
WHERE DLocation= 'Houston' AND DNo=DNumber
```

```
SELECT SSN, FName  
FROM EMPLOYEE  
WHERE EXISTS (  
    SELECT *  
    FROM DEPT_LOCATIONS  
    WHERE DLocation= 'Houston' AND DNo=DNumber )
```



Example 21

- Retrieve employees who have dependents with the same first name and gender as the employees

Example 19

- Find employees who have no dependents

Example 22

- Find employees whose salary is greater than at least one salary of employees in the department 4

Example 24

- Find managers who have at least one dependent

Discussion – IN and EXISTS

■ IN

- <Column_name> IN <Subquery>
- Attributes in the subquery's SELECT clause have the same data types as attributes in the outer query's WHERE clause

■ EXISTS

- Do not need attributes, constants or any expressions before it
- Do not need to specify column names in the subquery's SELECT clause
- Queries containing “= ANY” or IN can be converted queries containing EXISTS

Example 17 – TRC

- Find employees (FNAME) who work on projects or have dependents

Example 18 – TRC

- Retrieve the FNAME of employees who participate in projects and have dependents

Example 19 - DRC

- Find employees (SSN, LNAME, FNAME) who have no dependents

Example 19' – TRC

- Find the FNAME of employees who work on projects and have no dependents

Next lecture

■ Content

- Advanced query
 - Division & group by

■ Read

- Chapter 5 [The complete book, 2E]
- Chapter 6, 8 [Fundamentals of Database System, 7E]

■ Watch

- Video clip

